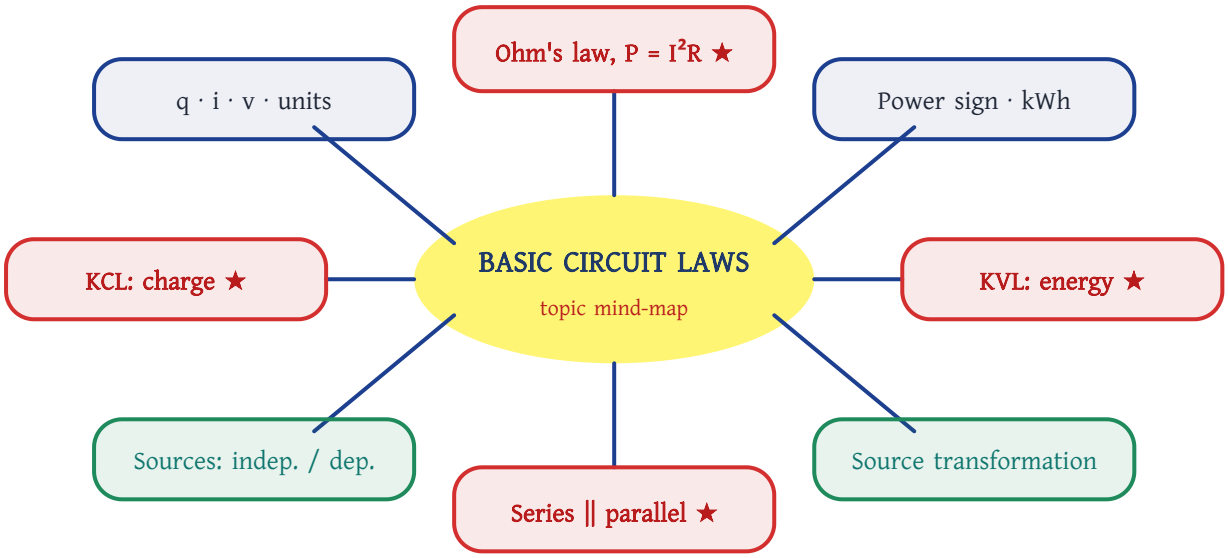
Circuit Variables & Basic Laws

V · I · P · Sources · Ohm · KCL · KVL · Dividers: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = method / applied

Syllabus coverage

"Circuit variables and elements: **voltage, current, power, energy**, **independent and dependent sources**, resistance. Basic laws of electrical circuits: **Ohm's law, KCL and KVL**. Simple resistive circuits: series and parallel circuits, **voltage and current division**, source transformation." Nodal/mesh analysis and the theorems (Thevenin, Norton, superposition, maximum power, Y–Δ) are in lesson 2; capacitors, inductors and AC are in lesson 3.

High-yield list

① $V = IR$, $P = I^2R$
② KCL: charge · KVL: energy
③ ideal V: $R = 0$ · I: $R = \infty$
④ $R_{parallel}$ < smallest R
⑤ divider rules
⑥ bulbs in series
⑦ $i_s = v_s/R$

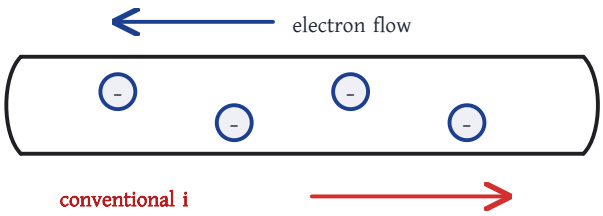
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

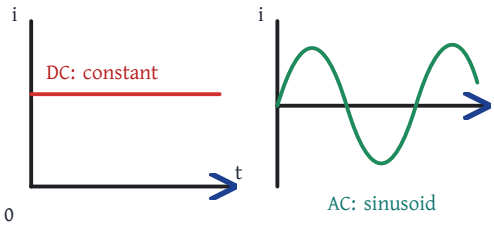
✓ Charge, current, voltage ..... p. 2
✓ Power, energy, sign convention ... p. 3
✓ Elements & sources ..... p. 4
✓ Resistance & Ohm's law ..... p. 5
✓ Nodes, loops & KCL ..... p. 6
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1 Charge & Current

Definition: **Charge** q is quantised: q = ne, with **e = 1.602 × 10^{-19} C**, so 1 C ≈ 6.24 × 10^{18} electrons. **Current** is the **rate of flow of charge: i = dq/dt**, and q = ∫ i dt. 1 A = 1 C/s.



conventional i = + charge direction · electron drift ≈ mm/s



DC vs AC (AC is lesson 3)

Quantity	Symbol	SI unit	In other units
Charge	q, Q	coulomb (C)	A·s
Current	i, I	**ampere (A)** ★ base unit	C/s
Voltage	v, V	volt (V)	J/C = W/A
Power	p, P	watt (W)	J/s = V·A
Energy	w, W	joule (J)	W·s = N·m
Resistance / conductance	R / G	ohm (Ω) / siemens (S)	V/A · A/V (S = ℧, mho)

2 Voltage (Potential Difference)

Definition: v_{ab} = **dw/dq**: the **energy needed to move a unit charge** from b to a. 1 V = 1 J/C. The + sign marks the higher-potential terminal, and $v_{ab} = -v_{ba}$.

Memory trick: Water pipe: **voltage = pressure**, **current = flow rate**, **resistance = narrowness** of the pipe. No pressure difference → no flow. Voltage is *across*, current is *through*.

Worked examples ✍️:

① 120 C in 1 min → I = 120/60 = **2 A**

② 3.2 mA → $3.2\times10^{-3}/1.6\times10^{-19}$ = **2×10^{16}** e/s

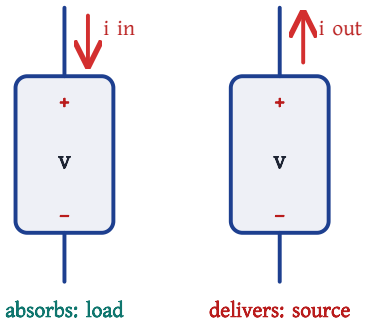
③ q = $10(1 - e^{-5t})$ mC → i = $50e^{-5t}$ mA; i(0) = **50 mA**

④ 30 J moves 2 C from b to a → v_{ab} = **15 V**

Exam trap: The SI **base** unit is the **ampere**; the coulomb is derived (A·s). Current has a direction but is a **scalar**: currents meeting at a node add algebraically, not as vectors.

3 Power & the Passive Sign Convention

Definition: Power is the **rate of energy transfer**: $p = dw/dt = v \cdot i$ (watt). **Passive sign convention:** if the current **enters the + terminal**, p = +vi is the power **absorbed**; if it enters the − terminal, use p = −vi.



v	i (enters at)	p absorbed	Meaning
5 V	2 A (+)	+10 W	load
5 V	2 A (−)	**−10 W**	gives 10 W
12 V	3 A (+)	+36 W	charging

Power balance: in any circuit **Σ p = 0**: total power supplied = total power absorbed (Tellegen's theorem).

Memory trick: "**In at plus = it eats**" (absorbs, a load). "**Out at plus = it feeds**" (delivers, a source). A resistor always eats: $p = i^2R \geq 0$.

4 Energy, kWh & Efficiency

Formulas:

w = ∫ p dt; constant P → **W = P · t**

→ **1 kWh = 3.6 × 10^6 J** = 1 "unit"

1 Wh = 3600 J · 1 hp = 746 W

$\eta = P_{out}/P_{in} \times 100\%$

Worked example 1 ✍: A 1.5 kW air conditioner runs 8 h a day for 30 days. W = 1.5 × 8 × 30 = **360 kWh** = 360 units; at an assumed Tk 8/unit the bill is **Tk 2880**.

Worked example 2: IT link ✍: Phone battery 5000 mAh at 3.7 V → 5 Ah × 3.7 V = **18.5 Wh** = 18.5 × 3600 = **66.6 kJ**. A 20 W charger at 85% efficiency gives 17 W → full charge ≈ 18.5/17 ≈ 1.09 h ≈ **65 min** (ideal).

Worked example 3 ✍: 12 V source, 4 Ω and 2 Ω in series: I = 12/6 = 2 A. Source gives 12 × 2 = **24 W**; resistors take 16 + 8 = **24 W** ✓ (power balance).

Exam trap: **kWh is energy, not power.** Do not confuse W (watt, the unit) with W (energy, the symbol). A battery being charged **absorbs** power: the current enters its + terminal.

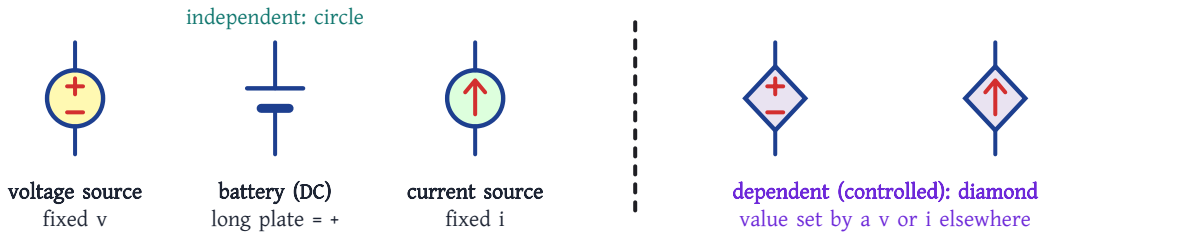
Also asked: 1 **electron-volt** = energy an electron gains through 1 V = **1.602 × 10^{-19} J** · a 60 W fan for 5 h uses 0.3 kWh · 1 J = 1 W for 1 s.

5 Circuit Elements: Active & Passive

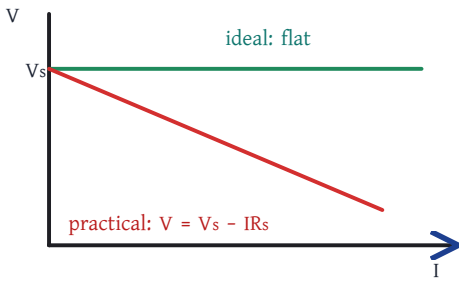
Definition: An **active element** can deliver net energy to the circuit (voltage and current sources: batteries, generators, op-amp outputs). A **passive element** cannot: the resistor **dissipates**, the inductor and capacitor only **store** energy (lesson 3).

Classification	Meaning (first / second)	Example
Linear / non-linear	V–I is a straight line through 0 / is not	resistor / diode
Bilateral / unilateral	same both directions / not	resistor / diode
Lumped / distributed	size ≪ wavelength / not	R, L, C / long line

6 Independent & Dependent Sources



Ideal source	Voltage	Current
Keeps fixed	v	i
The other	any i	any v
Internal R	0	∞
Never	short it	open it
Practical	V_s + series R_s	I_s ‖ R_p



terminal voltage falls with load current

Dependent source	Output	Gain (name, unit)	Models
VCVS	$v = \mu v_x$	voltage gain, **no unit**	op-amp
CCVS	$v = r i_x$	transresistance, **Ω**	transimpedance amplifier
VCCS	$i = g v_x$	transconductance, **S**	FET (g_m)
CCCS	$i = \beta i_x$	current gain, **no unit**	BJT (β)

Memory trick: Read the name **controller first, output last**: CC-VS = "current-controlled, it is a voltage source". Same kind in and out → no unit; mixed → Ω (v per i) or S (i per v).

Exam trap: Two ideal voltage sources of different values in **parallel**, or two different current sources in **series**, are impossible (they break KVL / KCL).

7 Resistance & Resistivity

Definition: Resistance is the opposition to current: **$R = \rho L/A$**, where ρ = resistivity (Ω·m), L = length, A = cross-section. **Conductance** G = 1/R in siemens (S, older ℧ "mho"); conductivity $\sigma = 1/\rho$ (S/m).

Material	ρ at 20 °C (Ω·m)
Silver (best)	1.6×10^{-8}
Copper	**1.7×10^{-8}**
Aluminium	2.8×10^{-8}
Nichrome (heaters)	1.1×10^{-6}

Temperature:

$R_t = R_0(1 + \alpha\,\Delta t)$

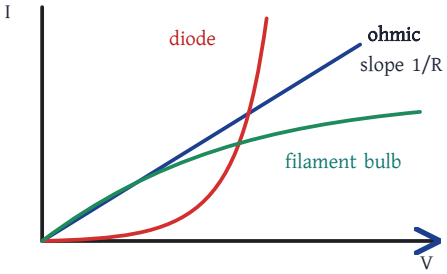
Metals: $\alpha > 0$ (Cu ≈ 0.0039 /°C)

Semiconductors, carbon, insulators, electrolytes: **$\alpha < 0$**

Manganin, constantan: $\alpha \approx 0$ → standard resistors

8 Ohm's Law

Statement: At **constant temperature and physical conditions**, the current through a conductor is directly proportional to the voltage across it: **V = I R** (G. S. Ohm, 1827). So I = V/R = GV and R = V/I.



straight line through 0 = ohmic

Power in a resistor (always absorbed): **$P = VI = I^2R = V^2/R$**

Short circuit: R = 0 → v = 0 for any i. **Open circuit**: R = ∞ → i = 0 for any v.

Non-ohmic: diodes, transistors, thermistors, arc lamps, electrolytes, a heating filament.

Worked examples ✍️:

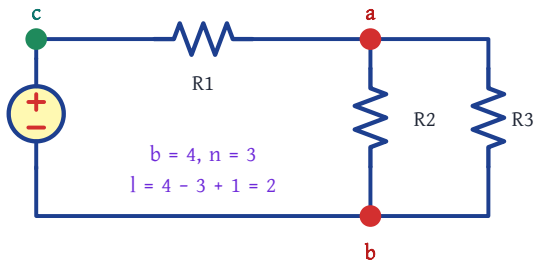
1. Cu, 100 m, 1 mm^2: $R = 1.72\times10^{-8} \times 100/10^{-6}$ = **1.72 Ω**; at 70 °C: ×(1 + 0.00393 × 50) ≈ **2.06 Ω**
2. Stretched to **2L**: volume fixed → A/2 → R × 2 × 2 = **4R** (n times → n^2R)
3. Bulb 100 W, 220 V: R = 484 Ω; on 110 V: $P = 110^2/484$ = **25 W** (½V → ¼P)
4. 12 V on 4 Ω → **3 A** · 2 A in 10 Ω → **40 W** · 1 kW, 220 V heater → **48.4 Ω**

Memory trick: **V-I-R triangle** (sheet 10): cover what you want. V = I×R, I = V/R, R = V/I.

Exam trap: Ohm's law is **not universal** (ohmic materials, constant temperature). KCL and KVL hold for **every** lumped circuit.

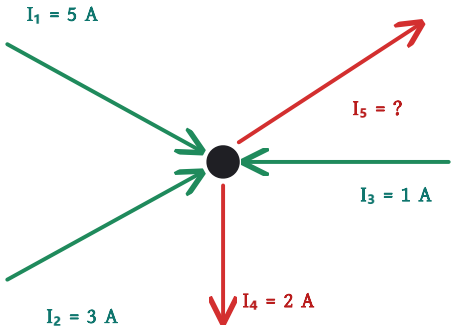
9 Nodes, Branches, Loops

- **Branch**: a single element (source or resistor) between two terminals
- **Node**: a point where **two or more branches meet**; all wire joined without an element is **one** node
- **Loop**: any closed path that visits no node twice
- **Mesh**: a loop with no other loop inside it
- Topology: **b = l + n − 1** (l = independent loops)



10 Kirchhoff's Current Law (KCL)

Statement: The **algebraic sum of currents at a node is zero**: Σ i = 0 (entering +, leaving −), i.e. **Σ i_{in} = Σ i_{out}**. Basis: **conservation of charge** (a node cannot store charge). G. Kirchhoff, 1845.



green = entering, red = leaving

Worked example 1 ✍️: In = out: 5 + 3 + 1 = 2 + I_5 → **I_5 = 7 A** (leaving).

Worked example 2 ✍️: 4 A and 2 A enter, 7 A and i_x leave: 4 + 2 = 7 + i_x → i_x = **−1 A**, i.e. 1 A actually **enters** through that branch.

Worked example 3: closed surface ✍️: KCL holds for any closed boundary (a **supernode**, a whole IC): three wires into a box carry 2 A in, 5 A in, so the third must carry **7 A out**, whatever is inside.

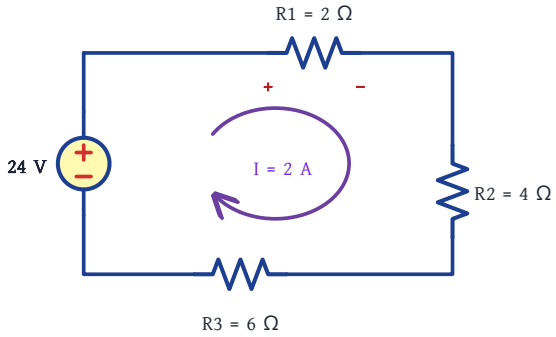
Consequences: Elements in **series** carry the **same current** · current sources in series must be equal · the current into a parallel group equals the sum of the branch currents · KCL at n − 1 nodes gives n − 1 independent equations (nodal analysis, lesson 2).

Memory trick: **KCL = Current = Charge** (three Cs). **KVL = Voltage = energy** (the "V" points down a hill of energy).

Exam trap: A **negative answer is not wrong**: it means the true direction is opposite to the one you assumed. Keep the sign, never "fix" it.

11 Kirchhoff's Voltage Law (KVL)

Statement: The **algebraic sum of all voltages around any closed loop is zero: Σ v = 0**, i.e. Σ voltage rises = Σ voltage drops. Basis: **conservation of energy** (a charge taken round a loop returns to the same potential).



walk round once, add every rise and drop

1. Mark polarities: current enters a resistor at its + end
2. Pick a direction and walk round the loop
3. Meet − first → **rise** (−v); meet + first → **drop** (+v)
4. Set the sum to 0 and solve

Worked example 1 ✍️: R = 2 + 4 + 6 = 12 Ω, I = 24/12 = **2 A.**
Drops: 4 V, 8 V, 12 V. KVL: −24 + 4 + 8 + 12 = **0** ✓
P: 8 + 16 + 24 = 48 W = 24 × 2 ✓

Worked example 2: opposing sources ✍️: A 12 V and a 4 V source face each other in one loop with 2 Ω + 2 Ω. Net emf = 12 − 4 = 8 V → I = 8/4 = **2 A**, flowing out of the 12 V + terminal and **into** the 4 V + terminal. 12 V source delivers 24 W; resistors take 8 + 8 = 16 W; the 4 V source **absorbs 8 W** (it is being charged) ✓.

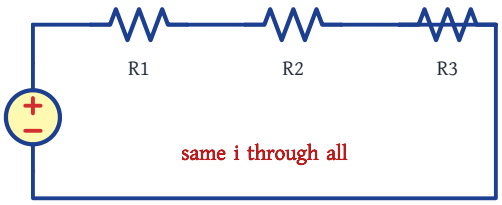
Worked example 3: voltage between two points ✍️: v_{ab} = sum of the drops along **any** path from a to b. In example 1, the voltage across R2 + R3 = 8 + 12 = **20 V**, and the other path gives 24 − 4 = 20 V ✓ (path-independent).

Consequences: Elements in **parallel** have the **same voltage** · series voltage sources add algebraically (aiding +, opposing −) · ideal voltage sources in parallel must be equal · KVL around the l = b − n + 1 meshes gives the mesh equations (lesson 2).

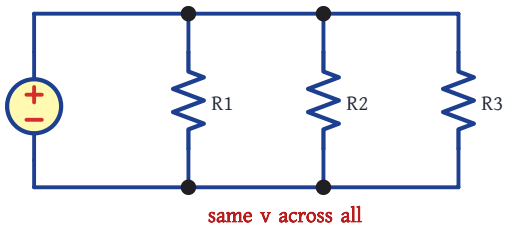
Validity: Both Kirchhoff laws hold for linear and non-linear, DC and AC (instantaneous or phasor), time-varying circuits, as long as the circuit is **lumped** (much smaller than the wavelength).

Exam trap: KVL adds voltages **with sign** (algebraic sum). Mixing the sign rule half-way round the loop is the most common written-exam error.

12 Series & Parallel Resistors



series: one path



parallel: shared pair of nodes

Feature	Series	Parallel
Common quantity	**current**	**voltage**
Divides	voltage ($\propto R$)	current ($\propto 1/R = G$)
Equivalent	$\mathbf{R_{eq} = R_1 + R_2 + \dots}$ (> largest R)	$\mathbf{1/R_{eq} = 1/R_1 + 1/R_2 + \dots}$ (< smallest R)
n equal R	nR	R/n (ratio series : parallel = $\mathbf{n^2}$)
Most power in	the **largest** R ($P = I^2R$)	the **smallest** R ($P = V^2/R$)
One R opens	everything stops	the others keep working
Examples	fuse, switch, ammeter	house wiring, voltmeter

13 Voltage & Current Division

Voltage divider (series):

→ $\mathbf{v_k = v \cdot R_k/R_{eq}}$

two R: $v_1 = v \cdot R_1/(R_1 + R_2)$

Current divider (parallel):

→ $\mathbf{i_1 = i \cdot R_2/(R_1 + R_2)}$

general: $i_k = i \cdot G_k/G_{eq}$

Worked example 1 ✍️: 15 V source, 3 Ω in series with (6 Ω || 3 Ω).

1. $6 \parallel 3 = 18/9 = 2\ \Omega \rightarrow R_{eq} = 3 + 2 = 5\ \Omega \rightarrow I = 15/5 = \mathbf{3\ A}$
2. v across the pair = $3 \times 2 = 6$ V (divider: $15 \times 2/5 = 6$ V ✓)
3. $I_{6\Omega} = 3 \times 3/9 = \mathbf{1\ A}$, $I_{3\Omega} = 3 \times 6/9 = \mathbf{2\ A}$; power $27 + 6 + 12 = 45\ W = 15 \times 3$ ✓

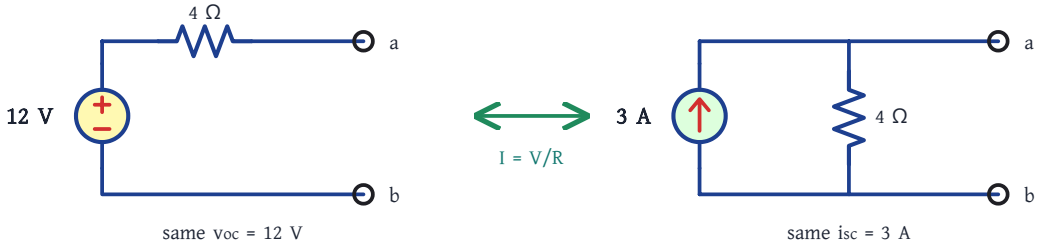
Worked example 2: bulbs ✍️: 100 W and 60 W (both 220 V) **in series** on 220 V: R = 484 Ω and 806.7 Ω, I = 220/1290.7 = 0.170 A → P_{100} = **14.1 W**, P_{60} = **23.4 W**: the **60 W bulb glows brighter**. In parallel each gets 220 V, so the 100 W bulb is brighter.

Memory trick: "**Series shares the current, splits the voltage; parallel shares the voltage, splits the current.**" Current divider: the **other** resistor goes on top.

Exam trap: $i_1 = i \cdot R_1/(R_1 + R_2)$ is **wrong**: the larger resistor takes the **smaller** current. A divider formula is only valid when no load draws current from the midpoint.

14 Source Transformation

Rule: A voltage source v_s **in series** with R is equivalent (at its terminals a–b) to a current source $\mathbf{i_s = v_s/R}$ **in parallel** with the **same R**, and vice versa. The current arrow points toward the + terminal.

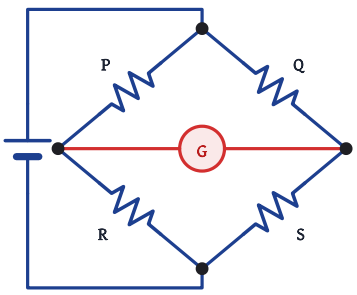


Worked example ✍️: Two practical sources, 12 V with 4 Ω and 6 V with 2 Ω, both feed a 4 Ω load in parallel. Find I_L.

1. Transform: 12/4 = 3 A ‖ 4 Ω and 6/2 = 3 A ‖ 2 Ω
2. Combine: 3 + 3 = 6 A ‖ (4 ‖ 2 = 4/3 Ω)
3. Back to a voltage source: 6 × 4/3 = 8 V in series with 4/3 Ω
4. $I_L = 8/(4/3 + 4) =$ **1.5 A**, $v_L = 6$ V (check with nodal analysis, lesson 2)

Exam trap: Impossible for an **ideal** source (R = 0 or ∞). An R in parallel with an ideal V source (or in series with an ideal I source) changes nothing at a–b.

15 Wheatstone Bridge



balanced: no current through G

Balance condition: **P/Q = R/S** (PS = QR). Then the galvanometer reads zero and the bridge needs no current measurement, only ratios.

Example ✍️: P = 10 Ω, Q = 20 Ω, R = 15 Ω → S = QR/P = 20 × 15/10 = **30 Ω**.

IT link: practical resistors: LED on a 5 V pin, $V_{LED} = 2$ V, 15 mA: R = (5 − 2)/0.015 = **200 Ω** → standard **220 Ω** (I ≈ 13.6 mA); $P = I^2R \approx 45$ mW, so a ¼ W part is ample.

Memory trick: colour code: "**BB ROY of Great Britain has a Very Good Wife**": Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9; gold ±5%, silver ±10%. Yellow-Violet-Red-Gold = 47 × 10^2 = **4.7 kΩ ±5%**.

Night-Before-the-Exam Sheet

Fact sheet:

$i = dq/dt \cdot v = dw/dq \cdot p = vi$

$e = 1.602 \times 10^{-19}$ C · 1 C ≈ 6.24×10^{18} e

$V = IR \cdot G = 1/R$ (S) · $R = \rho L/A$

$P = VI = I^2R = V^2/R$

1 kWh = 3.6 MJ = 1 unit · 1 hp = 746 W

$R_t = R_0(1 + \alpha\Delta t)$

KCL: $\Sigma\, i_{in} = \Sigma\, i_{out}$ (charge)

KVL: $\Sigma\, v = 0$ round a loop (energy)

Series: $R_{eq} = \Sigma R$ · same i

Parallel: $1/R_{eq} = \Sigma 1/R$ · same v

$v_k = v\, R_k/R_{eq}$

$i_1 = i\, R_2/(R_1 + R_2)$

$i_s = v_s/R$, same R

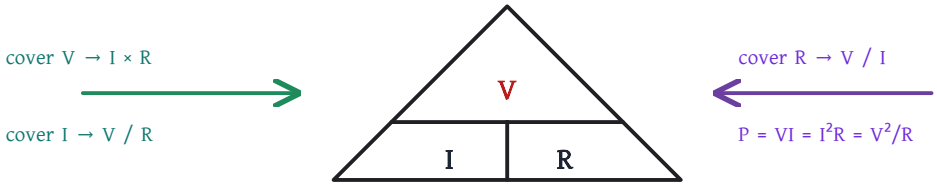
b = l + n − 1 · bridge: P/Q = R/S

Top 14 one-liners for MCQs

1. SI base unit: ampere (coulomb is derived)
2. Unit of conductance: siemens (mho, ℧)
3. KCL ← conservation of charge
4. KVL ← conservation of energy
5. Ideal voltage source: $R_{int} = 0$
6. Ideal current source: $R_{int} = \infty$
7. 6 Ω || 3 Ω = 2 Ω (product/sum)
8. Stretch wire to 2L → 4R
9. Half the voltage → quarter the power
10. Bulbs in series: low-watt one brighter
11. CCVS gain in Ω, VCCS gain in S
12. Ohm's law needs constant temperature
13. Silver is the best conductor; copper is used
14. Semiconductors: α negative

Top traps: kWh is energy, kW is power · negative current answer = opposite direction, not a mistake · current divider puts the **other** R on top · never short an ideal V source or open an ideal I source · no source transformation for ideal sources · Ohm's law is not universal, Kirchhoff's laws are (for lumped circuits) · current is a scalar

All tricks together: Water pipe: pressure, flow, narrowness · "in at plus = it eats" · controller first, output last (CC-VS) · V-I-R triangle · KCL = Current = Charge · series shares current, parallel shares voltage · BB ROY of Great Britain has a Very Good Wife



the V-I-R triangle: cover the quantity you want

✓ Practice MCQs (exam pattern, 10 marks)

1. The SI unit of conductance is:
(a) ohm (b) henry
(c) siemens (d) farad

2. 1 kWh equals:
(a) 3.6×10^6 J (b) 3.6×10^3 J
(c) 1000 J (d) 36 kJ

3. KVL is based on the conservation of:
(a) charge (b) energy
(c) momentum (d) mass

4. 4 Ω in series with (6 Ω ‖ 3 Ω) equals:
(a) 13 Ω (b) 6 Ω
(c) 2 Ω (d) 4.5 Ω

5. Internal R of an ideal current source:
(a) zero (b) equal to the load
(c) 1 Ω (d) infinite

6. Wire R stretched to 2× length; new R:
(a) R/2 (b) 2R
(c) 4R (d) R/4

7. 100 W, 60 W (220 V) bulbs in series:
(a) 100 W brighter (b) equal
(c) 60 W brighter (d) neither glows

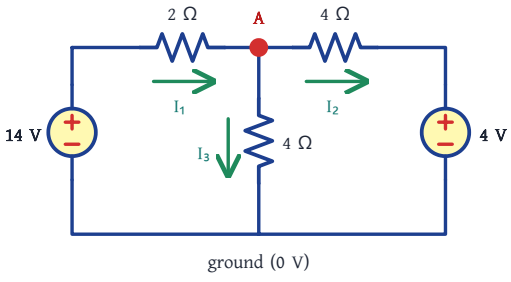
8. The gain of a CCVS is measured in:
(a) ohm (b) siemens
(c) no unit (d) ampere

9. 12 V in series with 4 Ω is equivalent to:
(a) 3 A ‖ 4 Ω (b) 48 A ‖ 4 Ω
(c) 3 A, series 4 Ω (d) 3 A ‖ 0.25 Ω

10. 8 A into 2 Ω ‖ 6 Ω; I in 2 Ω:
(a) 2 A (b) 8/3 A
(c) 4 A (d) 6 A

Answer key → 1-c · 2-a · 3-b · 4-b · 5-d · 6-c · 7-c · 8-a · 9-a · 10-d

Written question: State KCL and KVL. In the circuit, find I_1, I_2, I_3 and v_A using KCL and KVL, and show that the power balances. 8 marks



1. KCL: $\Sigma i_{in} = \Sigma i_{out}$ (charge); KVL: $\Sigma v = 0$ (energy) 1
2. KCL at A: $\mathbf{I_1 = I_2 + I_3}$ 1
3. KVL: $14 = 2I_1 + 4I_3$ and $4I_3 = 4I_2 + 4$ 2
4. $14 = 8I_2 + 6 \rightarrow \mathbf{I_2 = 1\ A}$, $I_3 = 2$ A, $I_1 = 3$ A, $v_A = \mathbf{8\ V}$ 2
5. 42 W given = 18 + 16 + 4 (R) + 4 (4 V absorbs) ✓ 2

Before the exam ✓

- ✓ Ohm, KCL, KVL with their basis
- ✓ Passive sign convention
- ✓ 4 dependent sources, gain units

Also practise (written):
① 2, 4, 6 Ω on 24 V: verify KVL
② Prove the current divider
③ Two-source source transform

✓ Series–parallel, both dividers